Top Open Source Tools *for* Network Security: The Old and the New

Data security vulnerabilities can result in tremendous losses, damage to reputation, and even national security threats. In this article, we discuss the top open source tools that will help systems administrators to enhance network security. A few of these are time tested, while others are new entrants trying to make a mark.

The past year has been a turbulent one for those in the cyber security domain, with events ranging from large scale malware attacks to the improper use of voter data in political campaigns hogging the headlines. Privacy concerns have risen to the fore with the European Union coming down heavily on the unregulated and possibly unethical use of personal data with the introduction of the GDPR (General Data Protection Regulation). There have been cyber attacks galore, without major public incident, save for the 'small' issue of millions of computers attacked by WannaCry and its successor, Petya.

Of course, the major leaks in the past year included parts of the CIA and NSA toolkits, and strategies to employ vulnerability brokers – overall, the year exposed the dark and murky secrets employed by the top intelligence agencies in their all-out effort to gain an edge in counter-terrorism efforts.

On the corporate front, fresh from the Cambridge Analytica scandal, Facebook announced a horde of changes targeted at alleviating user concerns regarding privacy. Amid all the furore over data, privacy and ethics, numerous tools have been introduced to address cyber security concerns. In this article, we delve into the list of the top open source tools for network security and a few new entrants.

The direction of growth

The past few years have seen the world plagued by large scale password leaks, DDoS attacks on widely used code hosting websites and, most recently, the accidental exposure of plaintext passwords by leading social networks. The recent focus on ethics and policy regarding data usage has seen much debate over the very definition of ethics. Overall, the industry seems to be growing more responsive to user concerns as most companies are falling in line with the GDPR, which has been ratified within the European Union. While the intrusion detection and security markets are largely catered to by the likes of proprietary offerings like McAfee, Symantec and Juniper, various open source variants are also being deployed within a large number of corporates. Intrusion prevention and detection has been the major focus in the launching of such tools. Let's look at what's on offer under the following two categories–the good old legends of network security and the newcomers finding a foothold in the industry.

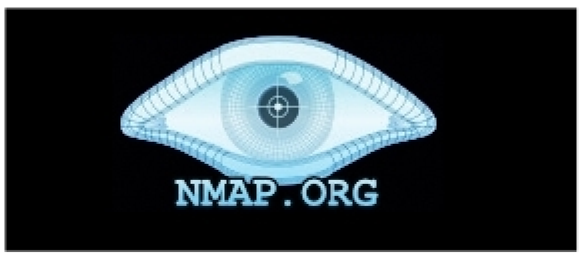

Figure 1: The NMAP tool (Credits: *nmap.org*)

The stalwarts

NMAP: Possibly the most popular port scanning tool of all time, NMAP has been going strong with an active community to back its development and adoption across the world. It is used to scan and map the network and various ports, and is backed by a powerful set of NSE scripts that can be employed to test and detect misconfigurations and security issues on the network. NMAP also sports its own version of Netcat, which is touted to be more powerful than the original. It has seen the addition of OS fingerprinting features and the expansion of the NSE scripts, among other performance improvements.

The Metasploit Framework: Impossible to miss out on, the Metasploit Framework focuses on defence from the view of the attacker. It offers a toolkit tailored to aid the security team in aggressively testing its own system for vulnerabilities —to perform security audits, and generate reports and assessments. The software comprises an arsenal of tools with contributions by experienced penetration testers in order to arm the defenders of a system against the

Figure 2: Rapid7's Metasploit Framework (Credits: *metasploit.com*)